

JharSetu

"Communities raise it. Campuses solve it. Industry scales it."

AI-assisted challenge exchange: citizen voice → compiled brief → verified → matched → funded → delivered with evidence.



Team: Computer Smashers · Platform: JharSetu v1.0

Build: Next.js 16 + Supabase + Gemini · Region: Jharkhand (24 districts) + Rajkot

Playbook v5 · September 2026

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01 What We're Building

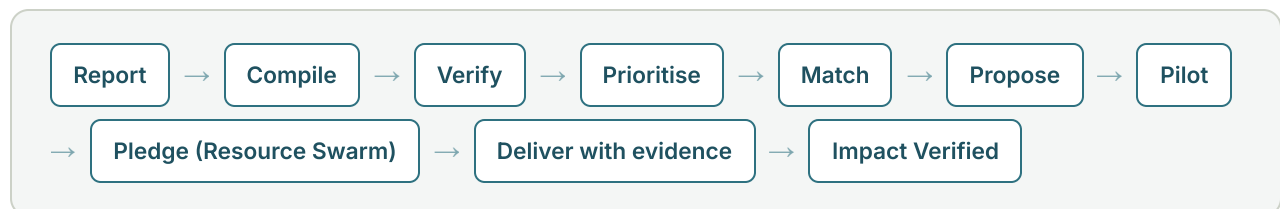
Platform for **Govt. of Jharkhand, SIH26043 (Disaster Management theme, 24h build, 100 marks)**. Not a complaint portal. A **challenge exchange**: villager voice note → verified, funded, staffed, tracked project with evidence.

LANDING HEADLINE

"See a need. Form a team. Close the loop."

Every university becomes the R&D department of its district.

The Full Loop



Two Operational Modes

PEACE MODE (DEFAULT)

Long-term challenges: education, health, farming, preparedness. Standard workflow, proposal competition, pilot phases.

CRISIS MODE

Triggered by real IMD/CAP alert or mock drill. Red top band + red districts + crisis room (needs, gaps, nearby resources, SMS reports, tech-squad call-up). Fast path: OPEN → DEPLOYED.

Geographic Scope

Region-first for **Jharkhand (24 districts)**, with live **region switch to Rajkot** (college round). Every seeded number labeled "*Simulated scenario*".

What This Is — and Is Not

✓ THIS IS	✗ NOT THIS
A challenge exchange: voice → verified → funded → tracked	A complaint portal (no feedback loop, no evidence, no closure)
University–industry collaboration platform with NEP 2020 credit	An emergency dispatch system (that's 112's job)
Evidence-gated closure (photos + beneficiary count + sign-off)	A disaster prediction AI
Hash-chained Impact Ledger for audit + CSR/NAAC PDFs	A general chatbot or payments system

02 Three Pitch Ideas

These are the three demo beats that must land in every presentation:

PITCH IDEA 1 — CHALLENGE COMPILER

Hindi/English voice/photo/text/SMS → structured brief + confidence score + *"AI is unsure about..."* + human approval gate. One call in, one brief out, every step timed and shown in the trace panel.

PITCH IDEA 2 — MATCHING + RESOURCE SWARM

List what's already nearby first (PostGIS radius query), then recommend university + company + NGO with reasons scored on 5 factors. Multiple partial pledges from different orgs fill a live gap bar. 8 pledges + 4 pledges = gap closes on stage.

PITCH IDEA 3 — IMPACT LEDGER

Closes *only* with photos + beneficiary count + volunteer check + community sign-off. Every action is hash-chained. Auto-generates CSR/NAAC/NIRF accreditation PDFs. Ledger can be independently verified at `GET /api/ledger/verify`.

Three Demo Beats (repeat everywhere)

- **Readiness Score 48 vs 84** — cheap deployable wins over impressive-but-impractical
- **Crisis drill + live SMS** — airplane mode on phone, SMS arrives with exact pin on map
- **Silent zones** — high hazard + real population + almost no reports = digital divide visualised

03 Roles & Demo Switcher

Six seeded accounts — one per role. One-click role switcher, no passwords on stage. Roles are real Supabase sessions enforced by PostgreSQL RLS, not hidden buttons.

CITIZEN

Report in <1 min. Track status. Confirm fix. No account required for reporting. Name and number never published.

FIELD VOLUNTEER / NGO

Field verify with photo + sources. List resources in registry. Confirm work on the ground.

COORDINATOR

District officer / faculty. Approve briefs. Rate readiness (1–5). Approve pilots. Close with evidence. Start drill. Override AI ranking.

UNIVERSITY

Adopt challenge. Build team (skill-gap plan). Submit PDF proposal. Execute pilot stages. Post progress updates.

INDUSTRY / COMPANY

Pledge materials (partial allowed). Get auto-generated CSR PDF. Message the college. Track delivery.

ADMIN

Verify orgs. Manage regions. Edit formula weights (logged). Award competition windows early. Full history of any challenge.

NGO — Separate Role (Decision §8)

ARCHITECTURE DECISION

NGOs are their own role (migration `0013ngo_role.sql`), separate from `industry` . Companies give materials; NGOs give money. They see different default boards.

What the Role Switcher Shows (RLS Proof)

ROLE	SEES PHONE/LOCATION?	CAN ADOPT?	CAN APPROVE?	DEMO LOGIN BODY
citizen	Own reports only	No	No	<code>{"role": "citizen"}</code>
volunteer	Fuzzed ~500m	No	No	<code>{"role": "volunteer"}</code>
coordinator	Yes (full)	No	Yes	<code>{"role": "coordinator"}</code>
university	No phone/location	Yes (verified)	No	<code>{"role": "university"}</code>
industry	No	Pledge only	No	<code>{"role": "industry"}</code>
ngo	No	Fund only	No	<code>{"role": "ngo"}</code>
admin	Yes	Yes	Yes	<code>{"role": "admin"}</code>

04 Technology Stack (Locked)

FRONTEND

Next.js 16 (App Router)

TypeScript

Tailwind CSS v4

shadcn/ui

Lucide React icons

PWA (installable)

Service Worker + IndexedDB offline queue

BACKEND (PORT 3001)

Next.js 15 (App Router, server-only)

Supabase (Postgres + Auth + RLS + Storage + Realtime + pgvector + PostGIS)

23 API route groups · 50+ endpoints

14+ DB migrations

Zod validation

Anthropic SDK + Gemini SDK

AI / LLM (WITH FALLBACKS)

Gemini 3.8/3.7/3.6/3.5-flash (cascade)

Anthropic Claude (secondary)

Groq / Whisper large-v3 (STT)

gemini-embedding-001 (768 dims)

Local hashing vectoriser (fallback)

Keyword classifier fallback

Rule-based Compiler (AI_ENABLED=false)

CORROBORATION (3 PROVIDERS)

Open-Meteo historical weather (free, unauthenticated)

News API / GNews / Newsdata (NEWS_API_KEY)

Tavily web search (SEARCH_API_KEY)

MAPS / CHARTS

Leaflet + OSM

leaflet.heat (heatmap)

Datameet district GeoJSON (Jharkhand + Rajkot)

[Recharts](#)[List-view fallback \(always\)](#)

DEPLOYMENT

[Vercel \(frontend: jharsetu-lilac.vercel.app\)](#)[Vercel \(backend: separate project, Root Dir = backend/\)](#)[Supabase \(ap-south-1 Mumbai\)](#)[SMS Gateway for Android \(spare phone\)](#)

Environment Variables Summary

VARIABLE	REQUIRED	WITHOUT IT
<code>NEXT_PUBLIC_SUPABASE_URL</code> + anon key	Yes	Platform does not start
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Yes	Reads fall back to anon (RLS blocks)
<code>GEMINI_API_KEY</code> (up to 10 keys rotate)	Strongly rec.	Rule-based compiler + no proposal scoring
<code>GROQ_API_KEY</code>	Strongly rec.	Voice notes save but don't transcribe
<code>NEWS_API_KEY</code>	Optional	Corroboration runs on weather + web only
<code>SEARCH_API_KEY</code> (Tavily)	Optional	Corroboration runs on weather + news only
<code>SMS_INBOUND_SECRET</code>	Optional	SMS webhook unguarded
<code>DEMO_PASSWORD</code>	Optional	Role switcher disabled
<code>AI_ENABLED=false</code>	—	Forces full fallback mode (demo kill-switch)

GOLDEN RULE

The **golden path must run with all AI keys removed**. The flow was built on the rule-based path first. `GET /api/health` shows what is wired up on each deployment.

05 Data Model (Supabase / PostgreSQL)

14 migrations applied: 0001_extensions → 0014_ngo_rls_parity. Every table carries `region_id`.

Core Tables

TABLE	KEY COLUMNS
regions	id, name, center, zoom, boundary, languages[]
users	id, role, org_id, region_id, language, district, skills[], reputation
organizations	id, region_id, type[univ company ngo govt volunteers], name, district, lat, lng, response_radius_km, csr_focus[], csr_budget, verified, embedding
resources	id, org_id, type, quantity, geom, availability
reports	id, client_id (idempotency), region_id, channel[web sms volunteer], phone_hash?, original_text, lang, audio_url, photo_urls[], geom, people_est, urgency, vulnerable[], translated_text, extracted jsonb, embedding, cluster_id, consent, created_at
challenges	id, region_id, title, brief jsonb, category, dm_phase, district, geom, people_est, severity, priority(0-100), score_breakdown, confidence, status, report_count, capabilities[], mode, crisis_id?, is_simulated
verifications	id, challenge_id, type, source_urls[], photo_paths[], note, verifier_id, created_at
matches	challenge_id, org_id, score, reasons[], status — write-through cache, recomputed on refresh
solutions	id, challenge_id, team_id, title, approach, cost_estimate, deploy_days, risks, readiness(0-100), status, proposal_pdf_path
resource_needs	id, solution_id, kind[money equipment people expertise], qty, unit, open_qty
pledges	id, need_id, org_id, qty, kind, status[pending sent received withdrawn]
ledger	id, entity, entity_id, action, actor, payload, prev_hash, hash, at — append-only hash chain
crisis_events	id, region_id, hazard, source, is_drill, districts[], severity, started, ended
notifications	id, user_id, challenge_id, type, message, read, created_at
progress_updates	id, challenge_id, stage_id (NOT NULL), note, photo_paths[], actor_id, created_at
proposal_competition_windows	id, challenge_id, opens_at, closes_at, winner_solution_id, status — migration 0010

Categories (8 fixed)

DISASTER/SAFETY

WATER

HEALTH

EDUCATION

AGRICULTURE

ROADS/INFRA

ENERGY/CONNECTIVITY

ENVIRONMENT

Hazard Tags + DM Phase

DM Phase (Sendai-aligned): `mitigation` | `preparedness` | `response` | `recovery`

Realtime Subscriptions (Supabase)

Enable Realtime in Supabase dashboard for: `challenges`, `pledges`, `resource_needs`, `notifications`. The gap bars and coordinator queue update without polling.

06 API Routes (23 Groups · 50+ Endpoints)

All responses: `{"ok":true,"data":{...}}` or `{"ok":false,"error":{"message":"...","code":"..."}}`. Frontend proxies `/api/*` via `next.config.ts` rewrite → same-origin cookie session.

Auth

POST	/api/auth/demo-login	Role switcher — body: {role}
GET	/api/auth/demo-login	Who am I now?

Reporting & Intake

POST	/api/reports	JSON or multipart (audio file). Idempotent on client_id. Returns trace, priority, confidence, decision. ports
GET	/api/reports/:id/trace	Transcribe→Translate→Extract→Dedup→Score→Draft with real timings (ms)
POST	/api/sms/inbound	Guarded by x-jharsetu-secret header

Challenges (Read)

GET	/api/challenges	filters: region_id, district, category, status, band, mode, org_id, q, limit, offset
GET	/api/challenges/:id	Full detail — everything the detail screen needs in one call
GET	/api/challenges/:id/solutions	
GET	/api/challenges/:id/actions	Drives the next-action button. Returns ready + problems.
GET	/api/challenges/:id/nearby	Registry resources within radius_km
GET	/api/challenges/:id/matches	?refresh=true recomputes + upserts cache
GET	/api/challenges/:id/gap	Live Resource Swarm bar
GET	/api/challenges/:id/timeline	Hash-chained ledger for this challenge
GET	/api/challenges/:id/similar	Do-not-duplicate library

Challenge Lifecycle (Write)

POST	/api/challenges/:id/approve	coordinator — returns full score_breakdown
POST	/api/challenges/:id/corroborate	AI check: weather + news + web, verdict + sources cited
POST	/api/challenges/:id/validate	field still_exists improved inaccurate more_affected unsuitable
POST	/api/challenges/:id/adopt	verified orgs only
POST	/api/challenges/:id/team	skill-gap team plan
POST	/api/challenges/:id/solutions	returns prefilled_ratings for reviewer
POST	/api/challenges/:id/pledges	partial pledge against one need
POST	/api/challenges/:id/evidence	before during after closure phases
POST	/api/challenges/:id/deploy	Evidence-gated closure. 422 with every unmet req if not ready.
POST	/api/challenges/:id/confirm	Community sign-off (the reporter)

Proposals & Competition

POST	/api/solutions/:id/review	7 readiness factors, each 1–5
POST	/api/solutions/:id/approve-pilot	creates needs + fires capability alerts
GET	/api/college/problems	university role — verified open challenges
POST	/api/college/proposals	multipart: challenge_id + document PDF
GET	/api/admin/windows	open competition windows
POST	/api/admin/windows/:id/award	award now (early close)

Crisis, Dashboard, System

POST	/api/crisis/start	{drill:true} or real IMD webhook
GET	/api/crisis/:id/room	needs board + gaps + nearby resources + tech squad
GET	/api/dashboard/metrics	KPIs, headline time-to-match, % reaching pilot
GET	/api/map/silent-zones	high hazard + real population + almost no reports
GET	/api/ledger/verify	recomputes every hash in the chain
GET	/api/health	what is wired up on this deployment

POST	/api/demo/reset	needs x-jharsetu-secret header
GET	/api/regions	region switch
POST	/api/admin/assistant	{question, challenge_ref?} — admin only
GET	/api/verify/queue	verifier, volunteer, coordinator, admin

07 The Five Formulas

All formulas are **deterministic code**. AI proposes inputs (severity, capability similarity); humans override. Admin-editable weights are logged in the ledger.

Formula 1 — Priority Score (0–100)

PRIORITY SCORE = SUM OF WEIGHTED FACTORS (TOTAL CAPPED AT 100)

- 25 pts** Severity (1–5). AI proposes; coordinator overrides.
- 15 pts** Urgency (deadline proximity + worsening rate)
Crisis Mode floors urgency at 80%. Inside 72h deadline it starts to dominate.
- 15 pts** People affected (log scale, saturates at 5,000)
- 15 pts** Vulnerability (children, elderly, disability, pregnancy, medical, isolation, no_signal)
Weighted individually (medical_dependency=1.0 → no_signal=0.5). Divisor = 3.4 (worst realistic combo).
- 10 pts** Hazard exposure (0–1 from GIS hazard_cells)
- 10 pts** Resource gap (% of need still unpledged)
- 5 pts** Recurrence (seasonal/repeated, saturates at 3×)
- 5 pts** Community signal (unique reporters, log scale, CAPPED)
Reporter count saturates fast — capped so loud urban areas cannot outrank remote villages.

= 100 Total

Bands: ≥75 CRITICAL 50–74 HIGH 25–49 MODERATE <25 LONG-TERM

Formula 2 — Confidence Ladder

Answers "is it real?" (separate from priority). Never hidden. An inaccurate flag drops one rung. AI never rejects.



Plain SMS from unknown number starts at bottom. External corroboration = AI checked Open-Meteo weather + news + web and cited every source. From `externally_corroborated` up, problems are open to colleges.

Formula 3 — Deduplication & Clustering

WITH SEMANTIC EMBEDDINGS (GEMINI)

cosine ≥ 0.85 AND ≤ 2 km AND ≤ 30 days $\rightarrow$ **merge**

0.75–0.85 AND same district $\rightarrow$ **review** (coordinator decides)

LEXICAL FALLBACK (NO EMBEDDING KEY)

same category AND ≤ 1 km AND ≤ 7 days $\rightarrow$ **merge**

same category AND ≤ 2 km $\rightarrow$ **review**

No-GPS report: same district + similarity $\geq 0.90 \rightarrow$ merge. A report with no GPS is not automatically a new problem.

Formula 4 — Capability Matching (0–100)

RESOURCE-FIRST: LIST REGISTRY ITEMS ALREADY IN RADIUS BEFORE PROPOSING NEW BUILDS

40 pts Capability fit (embedding similarity; fallback: keyword overlap)

20 pts Availability (free capacity now)

15 pts Proximity (decay half-life 35 km; unknown = 0.3)

15 pts Resource fit (% of open gap this partner could cover)

10 pts Partner type: univ (track record + interest) | company (CSR + district) | NGO (local presence + speed)

Every recommendation carries its reasons. Coordinator can override the ranking (override written to ledger).

Formula 5 — Solution Readiness Score (0–100)

EACH FACTOR RATED 1–5 BY COORDINATOR OR FACULTY MENTOR. COST/TIME/RESOURCES PRE-FILLED FROM PROPOSAL + REGISTRY.

- 20 pts** Technical feasibility
- 15 pts** Cost feasibility (cost $\leq$ ₹50k $\rightarrow$ 5; $\leq$ ₹2L $\rightarrow$ 4; $\leq$ ₹7.5L $\rightarrow$ 3; $\leq$ ₹25L $\rightarrow$ 2; else 1)
- 15 pts** Time to deploy ($\leq$ 30d $\rightarrow$ 5; $\leq$ 90d $\rightarrow$ 4; $\leq$ 180d $\rightarrow$ 3; $\leq$ 365d $\rightarrow$ 2; else 1)
- 15 pts** Local resource availability (from registry coverage %)
- 15 pts** Safety
- 10 pts** Community acceptance (from community validation signals)
- 10 pts** Scalability

THE DEMO BEAT THAT LANDS HARDEST

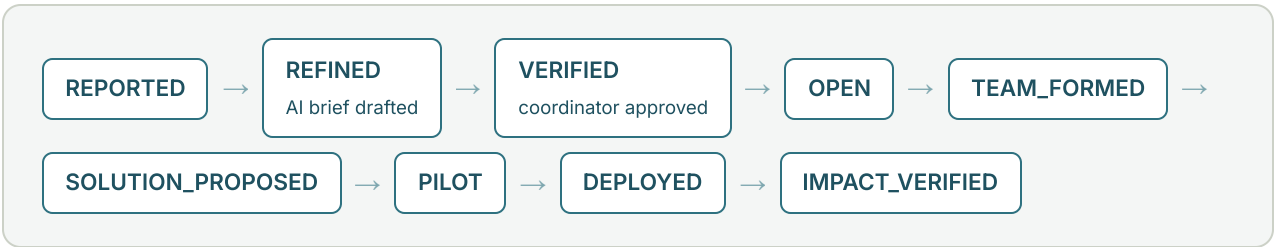
Sensor-AI prediction system: **48 / 100** (impressive but impractical).

Low-cost siren relay + shelter mapping: **84 / 100** (cheap + deployable next month).

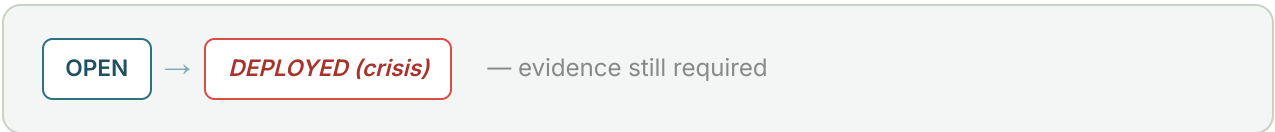
The 84 wins the pilot.

08 Lifecycle + State Machine

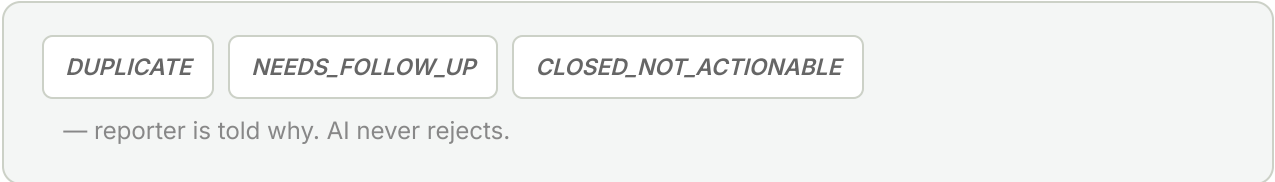
Main Path



Crisis Fast Path (crisisOnly)



Side Paths (human-only, reason required)



Transition Rules (from lifecycle.ts)

FROM	TO	WHO	REQUIREMENTS
REPORTED	REFINED	coordinator, admin	brief drafted
REFINED	VERIFIED	coordinator, admin	brief drafted
VERIFIED	OPEN	coordinator, admin	—
OPEN	TEAM_FORMED	partner roles	team formed
TEAM_FORMED	SOLUTION_PROPOSED	partner roles	proposal submitted
SOLUTION_PROPOSED	PILOT	coordinator, admin	pilot approved
PILOT	DEPLOYED	coordinator, admin	evidence + beneficiary count + verifier sign- off
DEPLOYED	IMPACT_VERIFIED	citizen, volunteer, coordinator, admin	community confirmation
OPEN / TEAM_FORMED	DEPLOYED	coordinator (crisis only)	evidence + count + sign-off

Evidence Requirements for Closure

- At least one photo or document as evidence
- Beneficiary count (number of people actually served)
- Field volunteer or coordinator verifier sign-off
- Community confirmation (`IMPACT_VERIFIED` stage)

HARD RULE

`POST /api/challenges/:id/deploy` returns HTTP 422 with every unmet requirement listed in `error.message` when any of the above is missing. The UI must render this verbatim.

Proposal Competition Window (Decision §9)

First proposal on a verified problem opens a 2–14 day window (by severity). Highest-scoring *viable* proposal wins when it closes. If none is viable, the window reopens for revision. Admins/coordinators can award early: `POST /api/admin/windows/:id/award`.

Auto-release

Claims are automatically released after 14 days of no progress. Reputation system tracks abandonment.

09 Frontend Screens (8 + sub-pages)

One primary next-action button per screen. Report works with no photo, no location, no account.

SCREEN / ROUTE	PRIMARY AUDIENCE	KEY COMPONENTS
Landing /	All	6-step how-it-works chain, CTA to /report and /login, public catalog links
Report a Need /report	Citizen	Voice recorder (Whisper STT), text/photo, district picker, SMS fallback link, offline queue, no account required
Challenges List /challenges	All	Filter by band/category/district/status, ChallengeRow with priority band, list+map views
Challenge Detail /challenge/:id	All roles	TraceSteps, ScoreFactors, Corroboration, Competition, ProjectWorkspace, Timeline, matches, Resource Swarm gap bar
Coordinator Dashboard /queue	Coordinator, admin	Approval queue, verify queue, active challenges, drill button, priority breakdown, one-click approve
Verify Queue /verify	Verifier, volunteer, coordinator	AI-corroborated items + manual verification with sources/photos
Project Workspace /projects	University, partner	Kanban: Ideas→Under review→Prototype→Field test→Deployed. Milestones, updates, evidence upload.
Needs Board /needs	Industry, NGO	Materials vs funding tabs, NeedCard with gap bar, pledge form, message thread
Silent Zones /silent-zones	Coordinator, admin, public	Map: high-hazard + real population + almost no reports. The digital divide visualised.
Crisis Room (triggered)	Coordinator, admin	Red top band, affected districts, needs board, gap, SMS reports with pin, tech-squad call-up
Impact / Overview /overview	All	KPI funnel, time-to-match, deployment rate, Recharts, region switch, metrics.simulated banner
Contributions /contributions	Industry, NGO	My pledges, tracked projects, message threads with college
My Reports /my-reports	Citizen	Reports I filed, current status, confirm fix button
Login / Role Switch /login	All	One-click demo role switcher. No password typed on stage.

Admin /admin

Admin

Org verification, weight editor, metrics, windows,
assistant, full history

Design Principle: Single Next-Action Button

Every screen has exactly one primary call-to-action that maps to the next legal lifecycle step for the current role. The button reads its state from `GET /api/challenges/:id/actions` .

10 Connectivity Ladder + SMS

1

Internet available

Normal upload — text + GPS location first, then audio/photo. Full trace panel. Realtime Supabase subscriptions.

2

No internet + signal + Critical (sev 4–5 or vulnerable groups)

Coded SMS via `sms:` link — one tap, pre-filled. Format: `JS1 K7F2 W5 25.2481,87.6412 P200 VCE "pani nahi 2 din"`

3

Basic phone (feature phone)

Plain Hindi SMS to the JharSetu number. Server parses and files. Confidence starts at bottom of ladder. Reply confirms receipt.

4

No signal (offline)

IndexedDB outbox queue + service worker retry. Idempotent `client_id` ensures no duplicate once connectivity restores. Volunteer assists on field if available.

Coded SMS Format (≤160 characters)

```
JS1 K7F2 W5 25.2481,87.6412 P200 VCE "pani nahi 2 din"
|  |  |  |
|  |  |  └── GPS coordinates
|  |  └── category + severity (W=water, 5=critical)
|  └── report ID
└── version 1
```

```
P200    = 200 people affected
VCE     = vulnerable: Children, Elderly [C/E/D/P/M/I/N]
"..." = brief description in local language
```

SMS REPLY

"JharSetu: report K7F2 received. For life-threatening emergencies, call 112."

Notes

- GPS only *receives* (incl. NavIC) — cannot transmit from browser; used to embed coordinates in SMS before sending
- Tower location + auto-SMS = native-app roadmap (browser cannot do this)
- Coded SMS lands with an exact pin on the crisis room map during the demo drill

- Production = licensed provider + TRAI DLT registration. Demo = spare Android + SMS Gateway for Android app
- Full sync later merges via the same report ID (idempotent)

Roadmap (Not Built — Say So)

- Missed call / IVR / USSD
- WhatsApp / Telegram (needs internet)
- Bhashini / Sarvam for Santali / Mundari / Ho (tribal languages)

11 Build Tiers — Strict Order

RULE

Golden path before extras. Working before pretty. Pretty before new. H18 = FREEZE (no new features, only fixes).

MUST — Golden Path (do first)

1. Auth + one-click role switcher
2. Report form (voice + text + photo + SMS link)
3. Challenge Compiler + rule-based fallback (AI_ENABLED=false must work)
4. Dedup (embedding + lexical fallback)
5. Confidence + priority + coordinator approve
6. Org / resource registry + nearby (PostGIS radius)
7. Matching + team builder (skill-gap)
8. Proposals + readiness score + pilot approval
9. Resource Swarm pledges + live gap bar
10. Lifecycle + evidence-gated closure
11. Impact record (beneficiary count + sign-off)
12. Map / list / dashboard
13. Notifications
14. Seed data + `POST /api/demo/reset` + cached AI responses

WOW — In Priority Order

1. **Crisis Mode** (red band, drill, fast path)
2. **SMS** (live coded SMS with exact pin on stage)
3. **Region switch** (Jharkhand → Rajkot in 10s)
4. **Silent zones** (hazard map + digital divide)
5. **Community checks + ledger + CSR/NAAC PDFs**

6. **Trace panel** (Transcribe→Translate→Extract→Dedup→Score→Draft with real ms timings)
7. **Offline queue** (IndexedDB + service worker)
8. **Do-not-duplicate library**
9. Solution co-pilot (only if all above are done)

PITCH ONLY — Never Build in 24h

Always say "roadmap" — never claim as built:

- Bhashini tribal language support (Santali/Mundari/Ho)
- Missed call / IVR / USSD / WhatsApp / Telegram
- Native iOS/Android app
- Report helper / ask-data chatbot
- Alert relay to other systems
- Weather satellite imagery
- DigiLocker / APAAR integration
- State relief portal integration

NEVER Build

- Rescue dispatch (that's 112's job)
- Disaster prediction
- Blockchain (hash-chain ledger instead)
- Payments
- General chatbot (3 focused assistants: compiler, co-pilot, admin only)
- Fake metrics or fake live feeds
- Screens off the golden path

12 Demo Script (8 Steps · 4:35 Total)

Skip the landing page. Start at /report. Two people can demo solo. Rehearse with slow-net + AI-down + mid-demo interrupt.

1 Report

0:30 – Gumla lightning, Hindi voice note, airplane-mode optional

Record a 15-second Hindi voice note about lightning strikes in Gumla. Submit with district, people count, and vulnerable groups. No English. No account.

"No form. No English. Works without network."

2 Compile

0:30 – Joins 31-report cluster, confidence rises, uncertainties shown

Trace panel shows: Transcribe (Whisper, 1.8s) → Translate → Extract → Dedup (cosine 0.91, 0.4 km from existing) → Score → Draft. Cluster count goes to 32.

"31 voices, one problem."

3 Approve

0:30 – Priority 82 breakdown + why critical

Coordinator sees: Priority 82 (Critical band), full factor breakdown, AI disclaimer, "AI is unsure about: exact tower count". One-click approve.

"AI prepares, human decides."

4 Match

0:30 – Siren stock Ranchi + halls, BIT Mesra ECE demo, fictional NGO + steel CSR

Nearby: Ranchi siren stockpile (12 km), community hall (8 km). Recommended: BIT Mesra ECE (score 87), fictional steel company (CSR match), local NGO. Team: 2 ECE + 1 CSE + 1 Civil + mentor.

"What's already nearby."

5 Propose

0:45 – 48 vs 84, pilot siren relay, 8+4 pledges close the gap

Two proposals: Sensor-AI (readiness 48) vs Siren relay (readiness 84). Pilot approved. College publishes needs: ₹80,000 + 20 siren units. 8 pledges from company (₹40k + 10 units) + 4 from NGO (₹40k). Gap bar closes live.

"Cheap next month beats clever next year."

6

Prove

0:35 – Seeded pilot evidence, volunteer check, Hindi notify, Somra confirms, ledger + CSR PDF

Seeded: installation photos + 200 beneficiaries. Volunteer sign-off. Reporter Somra Munda receives Hindi notification and confirms fix. Ledger entry verified. CSR PDF auto-generated.

"Closes with evidence."

7

See

0:30 – KPIs, funnel, silent zones, switch to Rajkot in 10s

Impact dashboard: time-to-match median, % reaching pilot, 150 JH challenges + 15 Rajkot. Silent zones map: Palamu block — high lightning hazard, 45,000 population, 2 reports. Switch region to Rajkot: waterlogging / heat / water / fire-prevention.

"Silence isn't safety."

8

Drill

0:45 – Mock IMD heavy-rain Sahebganj, red room, live SMS with data/WiFi off

Coordinator starts mock drill: Sahebganj heavy-rain, districts turn red. Turn data + WiFi off on demo phone. Type coded SMS to JharSetu number. Within seconds: exact pin appears on crisis room map with "via-SMS" badge. Env-lab accepts assignment. Buzzes tech-squad.

"Works when everything else fails."

Seed Data Reference

SCENARIO	DATA
Gumla lightning cluster	31 reports, 2 proposals (readiness 48 vs 84), 1 evidence-stage pilot, 200 beneficiaries
Sahebganj flood	1,250 people, 4 needs (water/200, medicine-route/350, connectivity, shelter), 2 SMS seeds, ledger entry
JH background	150 reports across 24 districts
Rajkot	15 reports (waterlogging/heat/water/fire-prevention — never use 2024 game-zone fire)
Orgs	10 universities (IIT-ISM/BIT/NIT/CUJ/BAU/Ranchi/RIMS — labeled "demo data, potential partner"), fictional companies/NGOs
Resources	40 resources in registry, 20 solved challenges

Non-Negotiables Before Stage

- Demo reset script works: `npm run db:reset` or `POST /api/demo/reset`
- Cached AI responses ready (network might fail)
- Backup SMS phone charged + connected to venue hotspot
- Screen recording of full 4:35 demo ready as backup
- 2 people can run the full demo solo if one has a problem
- Rehearse: slow-net mode + AI-down mode + mid-demo interrupt

13 Execution Checkpoints (24h Build)

Hour	Milestone
H0-1	Lock scope. Schema + region_id + client_id locked. API contracts agreed. Mock service stubs. Color tokens and SMS format defined.
H1-2	Design contract: wireframes for 8 screens. Supabase project created (Mumbai region). First migration applied.
H2-6	Foundation: all 14 migrations, seed data, report endpoint, compiler (rule-based fallback), dedup, priority. H6 checkpoint: report lands in deployed site feed. Auto-deploy working.
H6-12	Core path: matching, team builder, proposals, readiness, pledges, lifecycle, evidence-gated closure, impact record. H9: full golden path runs on fallback (no AI keys). H12: real AI integrated. Incognito run test.
H12-18	WOW features in order: Crisis Mode → SMS → Region switch → Silent zones → Ledger + PDFs → Trace panel. H18: FREEZE. No new features, only fixes.
H18-20	Test on production URL. Fix bugs found. Run smoke test: <code>npm run smoke</code>
H20-22	Deck finalized. Backup video recorded. Q&A drill with full team.
H22-24	Timed rehearsals. Submit. Rest the presenter.

Standup Cadence

Every 3 hours: What did I ship? What's blocking? Am I on the golden path? Stuck >30 min → say so immediately.

Lead Veto

Lead has scope veto at any checkpoint. If an extra feature risks the golden path, it gets cut. Small PRs, merge every 2–3h, one owner per task.

14 Scoring Rubric (100 Marks) + KPIs

PTS	CATEGORY	HOW TO WIN IT
10	Problem Statement	7-of-7 slide, Somra's story, lightning → 2,400 deaths over 10 years (2014–24)
15	Innovation	Comparison table, 48 vs 84 beat, live gap bar, mock drill
25	Technical Excellence	Trace timings shown live, architecture diagram, accuracy on 40 hand-labeled reports, live SMS with exact pin, AI kill-switch demo
10	UX / Usability	Phone report flow <1 min, 1-click approve for coordinator, Mukta font ≥16px
15	Feasibility / Impact / Scale	Rajkot region switch live, roadmap slide, CSR PDF shown, Sendai + NEP 2020 citations
5	Security / Privacy	Role switch hides PII (show RLS proof), fuzzed coordinates, SMS hash, DPDP mention
15	Presentation	Skip landing, finish 30s under

		time, confident Q&A
5	Teamwork	Clear roles shown, commit history, task board visible

Headline KPIs (show live on Impact dashboard)

- **Time: report → team formed** (median, hours)
- **% of challenges reaching pilot/deployment**
- **University–industry collaborations formed**

Speed / quality / outcome funnel. All seeded = simulated. Say so. Never claim cost-saved.

Key Phrases for Q&A

QUESTION	ONE-LINE ANSWER
Diff from grievance portal?	Continuous citizen sourcing + pilot to verified impact, not one-and-done complaint
Diff from 112?	We do hours-to-months: knowledge, equipment, funding. Never dispatch.
Why AI?	Seconds vs days for compile/dedup/match. Human decides. AI never rejects.
AI wrong or down?	Uncertainty shown + editable + % measured. Kill switch: AI_ENABLED=false. Rule-based fallback always runs.
Spam?	Confidence ladder + flags + dedup + EXIF strip + rate limits + human gate
No smartphone?	Plain SMS + volunteer assist + silent-zone surveys
Offline/GPS?	GPS receives only. SMS = low-bandwidth store-forward (Haiti 4636 precedent). IndexedDB queue.
Why Jharkhand from Rajkot?	Sponsor = Jharkhand. Region switch proves cross-state scalability in 10 seconds.
After hackathon?	1 district × 1 monsoon pilot. Measure the KPIs. CSR/grants fund it. State owns as infra.

15 Jharkhand Context & Policy Hooks

OPENING GREETING

"Johar!" — traditional greeting of the tribal communities of Jharkhand.

Why Jharkhand — The Numbers

- **Lightning: top killer** — >2,400 deaths over 10 years (2014–24), predominantly farmers in the open
- Jharia coal-seam fires: IIT-ISM Dhanbad nearest-capable university (demo example)
- Sahebganj Ganga floods: annual, affects riverside communities heavily
- Palamu / Garhwa drought and heat stress: increasing severity
- Elephant conflict: Singhbhum and Seraikela corridors

Tribal Languages (Roadmap)

Major languages beyond Hindi: **Santali, Mundari, Ho, Kurukh, Nagpuri, Khortha**. Voice + SMS + Bhashini integration = roadmap. Current: Hindi + English only. Say "roadmap".

Policy Alignment

POLICY / SCHEME	HOW JHARSETU ALIGNS
PM Disaster Risk Reduction 10-Point Agenda, Point 6	University network for local disaster solutions
Sendai Framework 4 Priorities	Mitigation, preparedness, response, recovery — all DM phases tagged
NEP 2020 Experiential Learning	Faculty mentor required for credit-bearing project work
UBA / NSS / Aapda Mitra	Verifier and volunteer roles in the platform
CSR Section 135 + Schedule VII	Disaster relief is explicitly listed. Auto-CSR PDFs.
DPDP 2023	Phone hash, location fuzz, consent, data minimisation, EXIF strip
Jharia Relief Portal (Dhanbad/Bokaro)	We complement, not replace. Different scope.

Authorities

NDMA → JSDMA (Jharkhand State Disaster Management Authority) → District DM authority. JharSetu works under district authority during crisis; never dispatches independently.

University Partners (Demo Data Only — Label "potential partner")

IIT-ISM Dhanbad, BIT Mesra, NIT Jamshedpur, Central University of Jharkhand, BAU Ranchi, Ranchi University, RIMS. All labeled "demo data, potential partner" in seed.

16 Design Tokens & Color System

Color Palette

TOKEN	HEX	USE
Ink	#102027	Primary text, dark backgrounds
Ground	#F4F6F5	Page/card background
Line	#CCD1C7	Borders, dividers
Teal	#2E7180	Actions, links, primary buttons. ONLY for actionable items.
Critical	#D94F45 / text #A8332A	Priority ≥75. Urgency ONLY. Never decorative.
High	#E07B2E / text #9A4A12	Priority 50–74. Urgency ONLY.
Moderate	#E5A83B / text #8A5A00	Priority 25–49. Urgency ONLY.
Long-term	#3867A6	Priority <25.
Resolved	#3E8064 / text #2F6B52	Verified/resolved status ONLY.

Color Rules

- **Never color-alone** — always pair with an icon + word for accessibility
- Red/orange/amber = urgency/severity bands ONLY
- Teal = actionable items (buttons, links) ONLY
- Green = verified or resolved status ONLY
- Crisis Mode = red top band across the entire application

Typography

FONT	USE	NOTES
Mukta	All body and UI text	Supports Latin + Devanagari. Minimum 16px. Line height 1.6. Touch targets 48px.
IBM Plex Mono	IDs, coordinates, timings, code, status chips	Used in trace panel, challenge refs (C-104), GPS coords, millisecond timings

AI Disclaimer — Show Verbatim

REQUIRED ON EVERY AI-ASSISTED OUTPUT

"AI-assisted recommendation. Final validation required from authorised coordinator."

Show why / data / uncertainty / verifier / next action. AI never dispatches, never rejects, never writes safety or medical advice (only verbatim NDMA/IMD cards + "Call 112").

17 Key Architectural Decisions (docs/DECISIONS.md)

Decision 1 — Backend is Source of Truth

The `backend/` app holds the intake pipeline, Challenge Compiler, dedup, priority, matching, readiness, lifecycle, ledger, RLS, and migrations. Frontend's `app/api/*` routes were a demo stand-in from hour 1 and are superseded. `backend/lib/seed/*` and `backend/scripts/seed-demo.ts` are canonical seed.

Decision 2 — One Origin, Not Two (Same-Origin Cookie)

`frontend/next.config.ts` rewrites `/api/*` to `BACKEND_ORIGIN`. The browser only ever talks to the frontend's own origin. Eliminates cross-origin cookie issues (Safari third-party cookie blocking, Chrome phase-out). CORS removed entirely for browser calls.

```
BACKEND_ORIGIN unset → demo routes answer (seeded data)
BACKEND_ORIGIN set   → real backend answers (no page changes)
```

Decision 3 — Reads via Service-Role Client

All reads go through `supabaseAdmin ?? supabase` (service-role). The `SUPABASE_SERVICE_ROLE_KEY` has no `NEXT_PUBLIC_` prefix and is only imported in `app/api/*` — never sent to browser. Anonymous `GET` on `challenges` would return empty (RLS blocks anon selects).

Decision 4 — Matches Table as Write-Through Cache

The `matches` table is a write-through cache. First call (or `?refresh=true`) invokes `computeMatches` and upserts results. Subsequent calls return instantly. Seed does NOT pre-populate: matches are computed against live database state, not seed time state.

Decision 5 — NGO as Separate Role

Migration `0013ngo_role.sql`. Companies give materials; NGOs give money. Different default boards, different RLS policies (`0014ngo_rls_parity.sql`).

Decision 6 — Proposal Competition Window

First proposal on a verified problem opens a 2–14 day window (by severity). Highest-scoring viable proposal wins. Reviewer marks not-viable with required changes. Window reopens if no viable proposal at close. Early award: `POST /api/admin/windows/:id/award`. Rule-based scoring if no AI key.

Decision 7 — Dedup Without Semantic Embeddings

With no embedding key, local hashing vectoriser scores differently-worded reports of the same flood around 0.4 (below 0.85/0.75 thresholds). Fallback: same category within 1 km in 7 days = merge. Within 2 km = review. Semantic embeddings keep the original bars.

Decision 8 — Progress Updates Belong to a Stage

`progress_updates.stage_id` is NOT NULL. General update with no stage → filed against the in-progress stage, else next undone stage, else last stage.

Decision 9 — End-to-End Test Script

`backend/scripts/e2e-flow.ts` : report → AI check → verifier → PDF proposal → award → requirements → 5kg + 5kg + money pledges → sent/received → messages → stage + update → admin record, metrics, assistant → public tracking, with role-wall checks. Rows titled `[E2E TEST]`. `--cleanup` deletes them. Ledger rows stay (append-only).

Vercel Deployment

- Frontend: root dir = `frontend/`, production branch = `main`, alias = `jharsetu-lilac.vercel.app`
- Backend: separate Vercel project, root dir = `backend/`
- Until GitHub auto-deploy is wired: `cd frontend && npx vercel --prod --yes`

18 Working Agreements for the Team

Core Rules

1. **Golden path before extras.** Working before pretty. Pretty before new. Freeze = no new features, only fixes.
2. **Never claim unbuilt as built** — use "roadmap". Every slide number needs a source or own measurement.
3. **Small PRs.** Merge every 2–3 hours. One owner per task. Don't edit others' files unasked.
4. **Stuck >30 minutes?** Say so immediately. Don't solo for 2h on a blocker.
5. **`.env` never committed.** All secrets in `.env.local`. See `.env.example`.
6. **All seeded numbers labeled "Simulated scenario".** Never display fake data without this label.

Locked Decisions — Don't Re-argue at Hour 14

DECISION	WHAT'S LOCKED
Name	JharSetu (not ResQGrid / AegisGrid / anything else)
Main problem	Lightning is the main killer demo (not flood, not drone delivery)
SMS vs messaging apps	SMS for offline/basic phone. No Telegram/WhatsApp.
AI chatbot	No general chatbot. Three focused assistants only (compiler, co-pilot, admin). Never built.
Region	Rajkot switch (not Jharkhand-only). Proves multi-region.
Stack	Next.js + Supabase + Gemini. No debate on switching frameworks.

Security Rules (enforced by RLS, not hidden buttons)

- Public map fuzz ~500m, strip photo EXIF publicly, hash SMS numbers
- Consent + anonymous option on every report
- DPDP 2023: minimise, purpose-limit, delete on request

- Zod + upload limits + rate limit (per SMS number)
- Orgs cannot adopt or pledge until admin-verified
- AI never dispatches, never rejects, never writes safety/medical advice

Dev Commands Quick Reference

```
# Backend (port 3001)
cd backend
npm install
npm run db:push          # apply all migrations
npm run db:seed          # seed both scenarios
npm run db:reset         # reset + seed (before each rehearsal)
npm run smoke            # judge's journey end-to-end test
AI_ENABLED=false npm run dev  # force rule-based fallback

# Frontend (port 3000)
cd frontend
npm install
npm run dev

# Full reset
npm run db:reset  # from repo root (runs backend scripts)
```

File Map

```
JharSetu/
  backend/
    app/api/**      23 route-group directories
    lib/ai/**        compiler, llm, fallback, corroborate, embeddings, stt
    lib/domain/**    priority, matching, dedup, confidence, readiness, lifecycle
    lib/services/**  intake, matching, swarm, crisis, ledger, notify, proposals
    lib/seed/**      data.ts + run.ts (Jharkhand + Rajkot scenarios)
    supabase/migrations/0001-0014 (extensions->ngo_rls_parity)
    scripts/         db-push, seed-demo, smoke, e2e-flow

  frontend/
    app/             All Next.js pages (15 screens)
    components/domain/ ChallengeRow, Competition, Corroboration, ProjectWorkspace...
    components/shell/ GovStrip, Logo
    components/ui/    Button, Card, Icon, States, Surface

  docs/
    API.md           Full endpoint reference
    BACKEND-SETUP.md Setup guide for new contributors
    DECISIONS.md     Architecture decision record (9 decisions)

  AGENTS.md         AI agent rules (source of truth for agents)
  context.pdf       This document
  README.md         Quick-start guide
```